The HCT Unit

4

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4.1 Introduction

Hematopoietic cell transplantation (HCT) is an advanced therapeutic intervention that is required for a number of malignant and nonmalignant medical conditions, often for critically ill patients. The establishment of an HCT program requires the efforts of experienced and appropriately trained personnel to lead the program. Clearly, this also requires financial, legal, ethical, and other institutional support. Without the commitment of the hospital director, allocation of resources, support of the national health authorities, and politicians, an HCT program will not be successful. For newly starting programs, it would be essential to identify minimal requirements for establishing an HCT unit in order to optimize resource utilization as well as maintain safe patient care. While these minimal requirements also apply to well-established units, its structure helps to understand and implement additional steps for larger units which plan to offer additional transplant services and have access to more resources. The recent advent of more cellular therapy types, including immune effector cell therapy, has added another layer of complexity necessitating additional requirements by HCT programs to ensure patient safety.

More than 20 years ago the EBMT and the ISCT (International Society of Cellular Therapy) formed the Joint Accreditation Committee—ISCT and EBMT (JACIE) based on the FACT (Foundation for the Accreditation of Cellular Therapy) program. Efforts of these bodies have culminated in the establishment of standards related to HCT and cellular therapies to assure quality and safety in the practice of HCT. Although program accreditation with JACIE is not mandatory worldwide, these standards are very helpful as guidelines to understand requirements to establish an HCT unit.

Table 4.1 summarizes basic minimal requirements of an HCT unit, which are discussed in more details in the following sections.

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Table 4.1 HCT unit minimal requirements

Inpatient unit	Clean single bedded rooms with isolation capability
Ancillary medical services	• Intensive care unit • Emergency room service • Gastroenterology and pulmonary service[a]
Outpatient clinic	Single-patient examination rooms
Blood bank	• 24 h on-site blood bank service: ABO typing and crossmatch, RBC and platelets for transfusion • Irradiation and leukocyte—depletion of blood products
Laboratory	• Hematology cell count and chemistry lab • Serology for viral screen • Microbiology for basic bacterial and fungal cultures • CMV PCR or antigenemia[a] • Access to CSA/tacrolimus levels[a]
HLA typing lab[a]	Access to ASHI or similarly accredited HLA typing lab
Stem cell collection	• PBSC apheresis capability • Bone marrow harvesting facility and expertise for matched sibling donor[a]
Stem cell processing facility	• FACS CD34 enumeration • Refrigerator for blood and bone marrow • Controlled cryopreservation capability for freezing of autologous stem cell product • Equipment and expertise to process ABO-mismatched cellular product[a]
Radiology	• Routine X-ray radiology, ultrasound, CT scanner • Placement of central venous catheters
Pharmacy	• Availability of conditioning chemotherapy drugs • Availability of antimicrobial agents (broad-spectrum antibiotics, antiviral and antifungal drugs) • Availability of immunosuppressive agents for GVHD prophylaxis and treatment[a]
Human resources	• Medical director: licensed physician with adequate training and experience in HCT • Nursing staff with training in chemotherapy administration, infection control, and handling of stem cell products • Clinical laboratory director—clinical pathology trained • Appropriately trained lab scientist and technicians • Clinical pharmacist trained in chemotherapy and conditioning regimen administration • Multidisciplinary medical staff (radiology, pathology, ICU, surgery, gastroenterology[a], pulmonary[a])
Outcome database	Monitor patient demographics, treatment, and outcomes (Level I data reporting)
Quality management	• Written institutional protocols/guidelines • Regular audits of various HCT procedures and patient treatment outcomes • System to detect errors or adverse events for corrective or preventative actions

[a] Requirements for allogeneic HCT programs

4.2 Inpatient Unit

The inpatient HCT unit should have a minimum number of single bedded rooms with isolation capability. The number and space of rooms should be adequate for the type and volume of transplant activity performed at the transplant center. These rooms must adhere to the standards of safety and comfort of patients in a tertiary care hospital facility. Every location or room should have a sink and tap for handwashing.

There needs to be a working station or room for nurses involved in patient care. A similar working space for physicians is required. Medical and nursing staff coverage should be available 24 h a day, including public holidays. The ratio of nurses to patient beds depends on the type and intensity of transplants being performed, e.g., autologous versus allogeneic, but generally a ratio of one nurse to three patients is reasonable. Emergency cart with drugs for resuscitation should be available in the inpatient unit.

Infections, including bacterial, viral, or fungal infections, are potential significant complications in transplant recipient and may lead to significant morbidity and mortality. Therefore, HCT units should have established measures for infection control. Guidelines for infection prevention and prophylaxis in HCT patients, endorsed by several scientific organizations, are available and highly recommended to follow. HCT recipients should be placed in single-patient rooms. Furthermore, at a minimum, standard precautions should be followed in all patients including hand hygiene, wearing of appropriate protective equipment (gloves, surgical masks or eye/face protection, gowns) during procedures/activities that are likely to generate splashes or spray of blood, body fluids, or secretions. Hand hygiene is essential, using alcohol-based hand rubs or washing with soap and water. In patients with suspected or proven of having an infection, additional precautions are required accordingly, e.g., airborne, droplet, or contact isolation. HCT units should be cleaned at least daily with special attention to dust control. During building construction intensified mold control measures are required and a multidisciplinary team (including infection control, engineers, infectious disease staff, and transplant clinicians) should be involved. Testing patients for multiresistant bacteria is important in order to isolate positive carriers and minimize transmission risk.

Other important infection control measures include well-sealed rooms, positive pressure differential between patient rooms and the hallway, self-closing doors, more than 12 air exchanges per hour, and continuous pressure monitoring. HEPA (high-efficiency particulate air) filters have shown efficacy in providing protection against acquisition of fungal infections in immune-compromised hematology patients, including HCT patients, and during hospital construction or renovation works. While HEPA filters are not absolutely required as a minimal requirement in newly established centers with less complicated transplant activities, they are certainly preferred and highly recommended as newly established centers expand their activities to include more complicated (especially allogeneic) transplant activities. HEPA filters are especially indicated in countries with high fungal burden, and therefore, HEPA filters are always preferred. Flowers and plants are not allowed in HCT units!

There is no agreed upon minimum number of transplants to be performed in a program. However, to ensure continuing proficiency in a transplant program, the ASBMT recommends for programs performing only one type of HCT (autologous or allogeneic), at least ten transplants of that type are to be performed per annum; programs performing both allogeneic and autologous transplantations should perform a minimum of ten transplants of each kind per annum.

4.3 Ancillary Medical Services

HCT patients often require other medical specialties' involvement in their complicated care. This include the risk of developing life-threatening infections or other posttransplant complications, hence the importance of having access to emergency room as well as intensive care services at the same tertiary care hospital facility where the transplant program is being established. Intensive care units should have the ability of providing inotropic support, respiratory support (including mechanical ventilation), and renal replacement (hemodialysis) if required.

Input from infectious disease physicians can be valuable in HCT patients who are at risk of a multitude of opportunistic and potentially life-threatening infections; this is especially important for programs that perform allogeneic transplants. Availability of gastroenterology specialist with endoscopy services is critical for allogeneic programs, as often diagnostic endoscopy is required to histologically differentiate GVHD from other etiologies of gastrointestinal complications. Similarly, pulmonary medicine service with access to diagnostic bronchoscopies is required for such patients with pulmonary abnormalities. Other ancillary medical services may include dermatology, ophthalmology, and gyne-

cology which can be valuable in the diagnosis and treatment of organ-specific GVHD complications in centers performing allogeneic HCT.

HCT programs that perform transplants using radiotherapy as part of a conditioning regimen (total body irradiation) should have available radiation oncology service on-site. The radiation oncology team, including the radiation oncologist and physicist, should have adequate training in the technique of TBI and appropriate equipment and procedures must be in place to deliver successful and safe radiation component of these conditioning regimen.

4.4 Outpatient Unit

HCT patients attend to the outpatient unit, both for pretransplant assessment and workup as well as posttransplant follow-up and management. Single-patient examination rooms are a minimal requirement for the outpatient service of the program. These rooms should be adequately equipped to allow clinical assessment of patients. It is important to implement infection control measures to minimize risk of transmitting infections, including hand hygiene measures and availability of appropriate room to isolate patients who are identified to be potentially infectious to others, e.g., due to herpes zoster infection. A dedicated infusion area would be ideal as transplant recipients often require IV fluid and electrolyte replacement or blood product administration.

4.5 Blood Bank

Availability of blood banking services is a critical component of a successful transplant program. A 24-h on-site blood banking service is required for ABO typing, crossmatch, and urgent supply of red blood cells and platelets for transfusion. Meeting minimal standard criteria according to recognized international blood bank societies such as the American Association for Blood Banking (AABB) or equivalent are important. Blood bank staff, including blood bank director, scientists, and technicians, should be adequately qualified and trained in blood banking procedures.

Transplant recipients are severely immune-compromised and are at risk of transfusion-associated GVHD, caused by unrestricted proliferation of donor lymphocytes in the immune-compromised host. Hence, it is critical that transplant recipients receive irradiated blood products to prevent this complication. The use of leukocyte-depleted blood products is recommended to reduce the risk of HLA alloimmunization in the multiply transfused hematology patients, as well as to reduce the incidence of transfusion reactions. In allogeneic programs, clear documented pathways for transfusion support in cases of ABO mismatch should be available for both blood bank and clinical staff as guidance.

4.6 Laboratory

A 24-h on-site hematology cell count and basic chemistry lab is required. Furthermore, microbiology laboratory service is essential in the clinical management of transplant recipients, including routine bacterial and fungal cultures of various patient specimens. Serology screening for relevant viral and bacterial infections is also required for pretransplant workup of recipients as well as donor screening, including testing for HIV, viral hepatitis, CMV, HSV, HTLV1, toxoplasma, and syphilis among others. For allogeneic transplant recipients, monitoring for cytomegalovirus (CMV) reactivation is essential and results must be available in a timely manner to allow therapeutic intervention; both molecular technique by quantitative PCR (preferable) and antigenemia method are acceptable. In the allogeneic setting, monitoring drug levels, e.g., cyclosporine or tacrolimus, is required, and same-day service is recommended to allow interventions aiming at keeping levels of these important drugs within the target therapeutic range.

4.7 HLA Typing Lab

Access to an HLA typing laboratory is mandatory for allogeneic programs. Such service can be available on-site or alternatively provided in reference laboratory. JACIE standards state that clinical programs performing allogeneic transplantation shall use HLA testing laboratories that are capable of carrying out DNA-based intermediate- and high-resolution HLA typing and are appropriately accredited by the American Society for Histocompatibility and Immunogenetics (ASHI), European Federation for Immunogenetics (EFI), or other accrediting organizations providing histocompatibility services appropriate for hematopoietic cellular therapy transplant patients.

4.8 Stem Cell Collection

Access to peripheral blood stem cell apheresis service on-site is a minimal requirement in each program. This is often part of the blood bank service or alternatively under the administration of the clinical program. Having at least two cell separators would be beneficial, as the second cell separator would be a backup in situations of unexpected machine faults and for routine servicing. Daily operation of apheresis facility requires appropriately trained and experienced nursing staff and a medical director with adequate qualification and experience in clinical and laboratory aspects of the apheresis procedure. Institutional written protocols and policies covering all aspects of apheresis procedure should be available for guidance. JACIE standards require a minimum average of ten cellular therapy products collected by apheresis per year for program accreditation.

A BM stem cell source is sometimes recommended for better patient outcome, e.g., patients with BM failure. Programs performing allogeneic HCT for such indication should have a bone marrow harvest facility on-site. This requires convenient and easy access to surgical operating room with anesthesia service. Appropriate equipment for the BM harvest procedure are required. Physicians with adequate training and experience in BM harvesting are crucial to perform the procedure successfully.

4.9 Stem Cell Processing Facility

The stem cell processing facility requires a designated area, usually within the laboratory. It should be appropriately equipped for the processing of various stem cell products depending on the types of transplants performed and the size of the program. Availability of flow cytometry for the enumeration of CD34 cell count is mandatory. Controlled cryopreservation capability, using liquid nitrogen, for freezing of autologous stem cell product is essential. This may also be used in allogeneic sibling products. Standard quality control measures, including systems to closely monitor and record the temperature in all freezers and refrigerators, are critical. Allogeneic programs should have appropriate equipment and expertise on-site for the timely and safe processing of ABO-mismatched stem cell products as required, including the need to perform red cell or plasma depletion procedures when indicated. The processing facility should be operated by adequately trained staff, including scientist, technicians, and a medical director. Written standard operating procedures explaining all aspects of stem cell processing performed at the facility are required.

4.10 Radiology

Standard routine (X-ray), ultrasound, and computed tomography (CT scan) imaging services are the minimal requirements and should be available on-site for the routine diagnostic imaging. Availability of magnetic resonance imaging (MRI) is preferred, as it is often useful in the diagnosis of specific clinical conditions relevant to stem cell transplant recipients, such as iron overload, CNS infections, and posterior reversible encephalopathy syndrome (PRES) related to

calcineurin inhibitor toxicity. Placement of central venous catheters in transplant recipients is obviously required in each program. Depending on the institutional setting, this service may be provided by various hospital services; often this is done by the radiology service under ultrasound guidance. Having well-trained and experienced interventional radiologist to perform this procedure is crucial for the safety of patients.

4.11 Pharmacy

Pharmacy services are essential in each HCT program. Availability of conditioning chemotherapy agents is clearly required; specific drugs depend on the type and complexity of transplant procedures performed in each program. Commonly used agents in conditioning regimens include busulfan, cyclophosphamide, fludarabine, and melphalan. Antithymocyte globulin (ATG) may also be required in the allogeneic setting (e.g., in aplastic anemia) and requires special attention and training by nursing, pharmaceutical, and medical staff in relation to its administration.

Broad-spectrum antibiotics should be available for urgent use as required in transplant recipient. Likewise, access to antiviral and antifungal agents is important for both prophylaxis and treatment. Allogeneic programs should also have access to immunosuppressive drugs used for GVHD prophylaxis such as cyclosporine, methotrexate, and tacrolimus, as well as medications required for GVHD treatment.

A trained pharmacist is crucial for the HCT program. The pharmacist should review all conditioning chemotherapy protocols and ensure appropriate dispensing and administration of cytotoxic agents.

4.12 Staffing and Human Resources

Appropriately trained and experienced medical and nursing staffs are crucial for the HCT program. The clinical medical director of the program should be a licensed physician (specialty certification in hematology, oncology, or immunology) with adequate training at a BMT program. A minimal BMT training duration of 6–12 months is suggested. JAICIE standards indicate that the clinical program director shall have 2 years of experience as an attending physician responsible for the direct clinical management of HCT patients in the inpatient and outpatient settings. A minimum of one (1) additional attending transplant physician is required in the program. Modern technologies like supervisory telemedicine may be useful to cope with lack of experience.

The success of a transplant program relies heavily on the presence of appropriately trained and experienced nursing staff. This includes training in chemotherapy administration, infection control, management of neutropenic patients, and handling of stem cell products.

Other important staff includes appropriately trained and experienced personnel in the laboratory (including laboratory director, scientist, and technicians), trained pharmacist, and medical professionals of ancillary medical services. Continuous education activities are required for all healthcare professionals involved in the management of HCT patients. Written HCT protocols are essential and should be signed by the physician and the pharmacist.

4.13 Chimeric Antigen Receptor T-Cell (CAR-T Cell) Unit

The use of this type of immune effector cell therapy is gradually expanding and is typically implemented in well-established HCT centers. CAR-T therapy represents a novel technology, not just in the production and cellular process that underpin the mechanism of action, but also in the processes, clinical pathways, and complications. Institutions, therefore, need to develop clear guidance for all healthcare providers who are involved in the CAR-T therapy process. All relevant staff involved in the prescribing, dispensing, or administering of CAR-T cell thera-

pies should be aware of how to manage the risks of cytokine release syndrome (CRS) and neurological complications and should be trained and certified in the respective manufacturers' risk evaluation and mitigation strategy (REMS) programs. The complexity and toxicity profile necessitate that CAR-T cell therapy is delivered in a center and a program that is experienced in the management of patients undergoing cellular therapy and its complications with close liaison and access to intensive care. Hence, the start of CAR-T cell therapy in HCT programs should be preceded by adequate preparation addressing multiple elements that include, but are not limited to, materials, equipment, personnel, documents (standard operating procedures and institutional guidelines), staff training, and facilities. On-site availability of relevant ancillary medical services (intensive care, neurology, and infectious disease), as well as medications required for the treatment of CRS and immune cell-associated neurotoxicity syndrome (ICANS), is paramount.

4.14 Institutional Database and Data Manager

Monitoring patient demographics, treatment details, and outcomes is an essential minimal requirement. Each program should keep complete and accurate patient records using unique patient numbers (UPN), and a database containing relevant patient data should be established and regularly maintained. Appropriate patient consent needs to be obtained for such database. An example of the minimal data required to be obtained on each transplant patient is the information required in the CIBMTR or EBMT MED-A forms. Having a data manager in a transplant program to initiate and maintain this institutional transplant database is highly recommended. Often data managers have a nursing background with experience in HCT. Attending training data management courses during international meetings or through links with other experienced and well-established programs would be valuable.

4.15 Quality Control

The JACIE standards require that all essential clinical, collection, and processing facilities in the transplant center evaluate and report patient outcomes. Regular audits of various HCT procedures and patient treatment outcomes are required. Essentially, a system is required to be in place to detect errors/adverse events, so that these can be evaluated in order to implement preventative measures to minimize the risk of recurrence of these incidents. Furthermore, each program should have written institutional clinical protocols in relation to the various aspects of the transplant patient care to standardize practice. Likewise, stem cell collection and processing facilities should have standard operating procedures that serve as a guidance for all staff to follow to enhance patient's safety. Access to or relationship with experienced HCT program is often very helpful and highly recommended via shared protocols/telemedicine and/or web-based conferencing.

4.16 Transplant Coordinator

HCT is a complex therapeutic intervention and coordination of the pretransplant, transplant, and posttransplant patient care is important. A transplant coordinator can play a pivotal role in this context, acting as a facilitator and educator as well as a point of contact for the patients and their families. Transplant coordinators ensure the smooth and safe running of the HCT service starting from scheduling and arranging pretransplant workup of patients and planning the roadmap for the transplant recipient with continued involvement and education of the patients and their families until the time of admission. Furthermore, the transplant coordinator would play a significant role in the coordination of post-HCT follow-up and care in clinics. For allogeneic HCT, the transplant coordinator would be very valuable in arranging donor search starting from HLA typing of the recipient and his/her family members, in addition to initiating and following a search for unrelated donor in national or interna-

tional registries. The transplant coordinator involvement may extend to organizing the logistics of getting the stem cells from the donor from the donor center where the recipient may be in another health facility (national or international). Moreover, transplant coordinators will often lead the HCT team's weekly planning meetings and discussions with the arrangement of the HCT waiting list. Typically, transplant coordinators have a nursing background with significant experience in stem cell transplantation.

Key Points

- The inpatient unit should have single bedded rooms with isolation capabilities. Single outpatient examination rooms are also required.
- Laboratory, blood bank, and pharmacy services are critical to the success of HCT programs.
- Stem cell collection and processing capabilities are minimal requirements for any HCT program; the level of such capabilities depends on the type and complexity of HCT performed in each center.
- Ancillary medical services are essential components of successful HCT programs, including intensive care, emergency, and radiology services. Additional medical services are required in allogeneic programs.
- Appropriately trained and experienced staff (medical, nursing, laboratory, pharmacy) are crucial for the HCT program.
- The complexity and toxicity profile necessitate that CAR-T therapy is delivered in a center and a program that is experienced in the management of patients undergoing cellular therapy and its complications with close liaison and access to intensive care. Hence, the start of CAR-T cell therapy in HCT programs should be preceded by adequate preparation addressing multiple elements, to mitigate risks associated with immune effector cell therapy.
- Monitoring patient characteristics and transplant outcomes is essential.
- A local quality control system is required in all aspects involved in the HCT procedure.
- Having a data manager for the HCT program, to initiate and maintain institutional minimal transplant database, is highly recommended.
- Transplant coordinators play a pivotal role in the management of HCT patients, starting from pre-SCT workup, right through posttransplant care.

Further Reading

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